

PENRITH HIGH SCHOOL



MATHEMATICS EXTENSION 2 2013

HSC Trial

Assessor: Mr Ferguson

General Instructions:

- Reading time – 5 minutes
- Working time – **3 hours**
- Write using black or blue pen. Black pen is preferred
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 11 – 16.
- Work on this question paper will not be marked.

Total marks – 100

SECTION 1 – Pages 2 - 7

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

SECTION 2 – Pages 8 - 13

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Section 1

Section 2

Question	Mark
1-3	
4-5	
6-7	
8	
9	
10	
Total	/10

Question	Mark		Question	Mark
11	/15		15 _{motion}	/5
12	/15		15 _{poly}	/7
13 _{vol}	/7		15 harder ext	/3
13 _{graph}	/8		16 _{complex}	/6
14 _{conic}	/12		16 harder ext	/9
14 _{circle}	/3			

Total	/100
%	

This paper MUST NOT be removed from the examination room

Student Name:

Section I

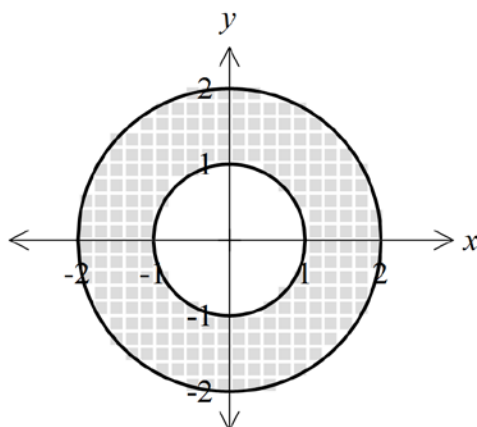
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Consider the Argand diagram below.

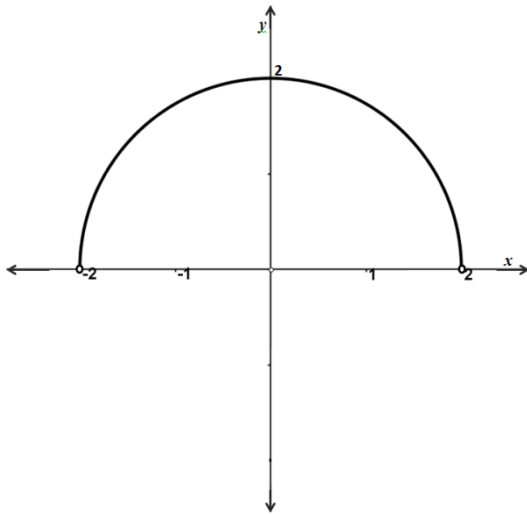


The inequality that represents the shaded area is:

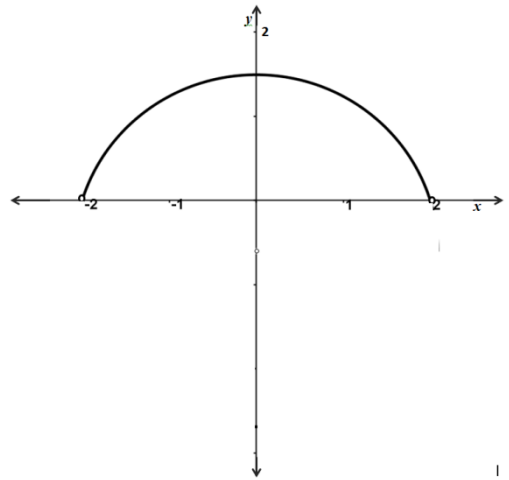
- (A) $0 \leq |z| \leq 2$
- (B) $1 \leq |z| \leq 2$
- (C) $0 \leq |z - 1| \leq 2$
- (D) $1 \leq |z - 1| \leq 2$
- 2 Let $z = 1 + 2i$ and $w = -2 + i$. The value of $\frac{5}{iw}$ is:
- (A) $-1 - 2i$
- (B) $-1 + 2i$
- (C) $1 - 2i$
- (D) $1 + 2i$

3 The locus of z if $\arg(z-2) - \arg(z+2) = \frac{\pi}{4}$ is best shown as:

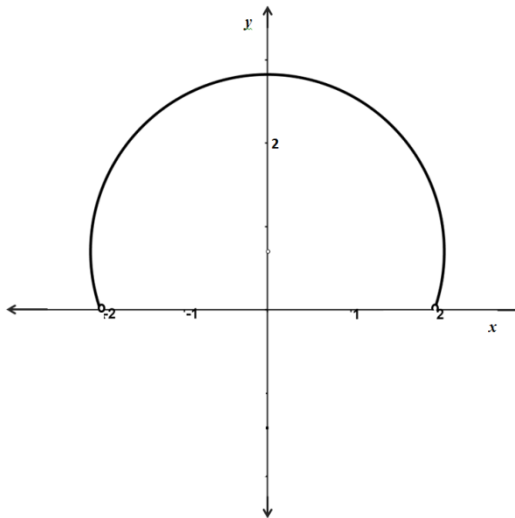
(A)



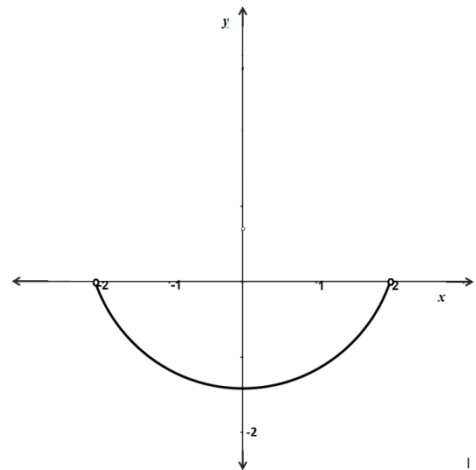
(B)



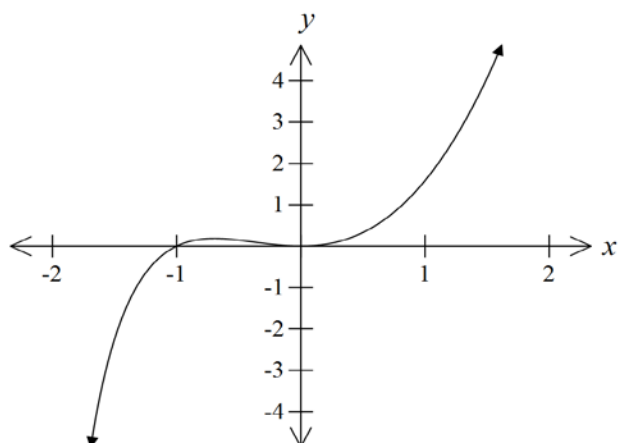
(C)



(D)

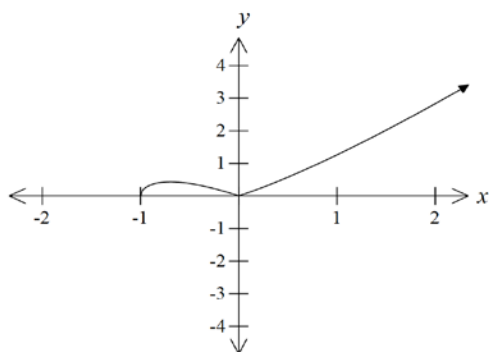


- 4 The diagram shows the graph of the function $y = f(x)$.

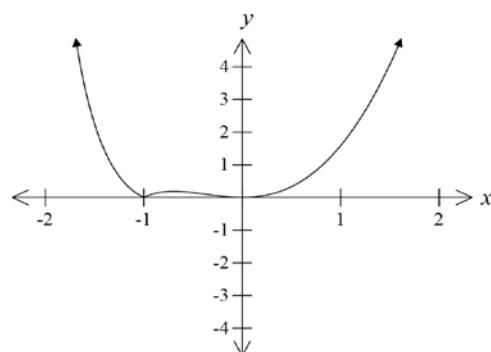


The diagram that shows the graph of the function $y = f(x)^2$ is:

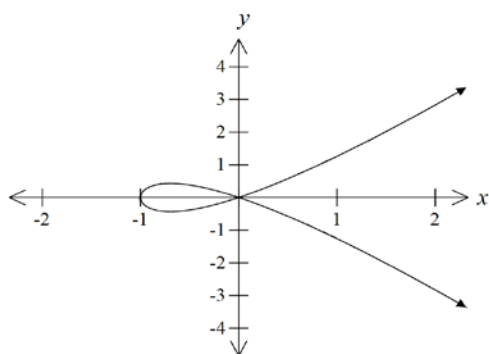
(A)



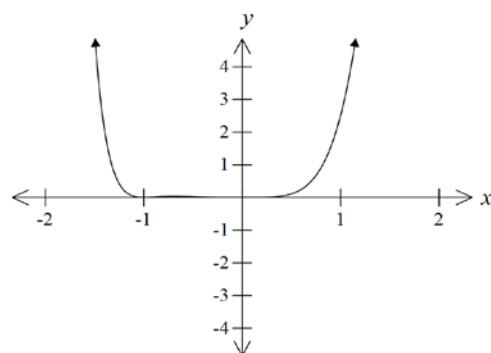
(B)



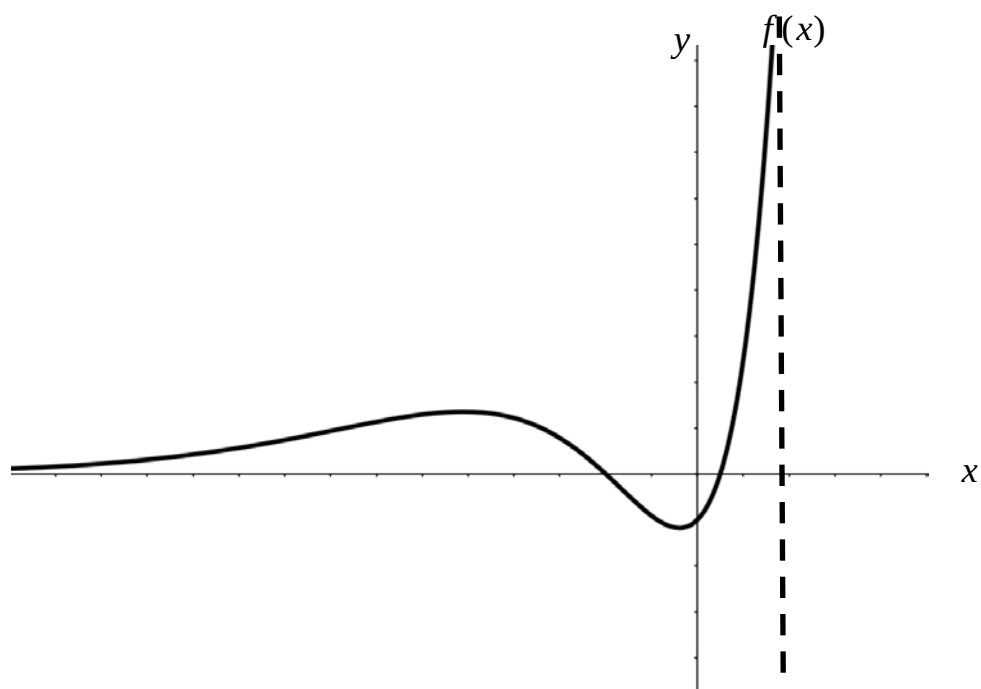
(C)



(D)



- 5 The function $y = f(x)$ is drawn below.



Which of the following is an incorrect statement?

- (A) $y = f(x)$ has two asymptotes only
 - (B) $y = f(x)$ is continuous everywhere in its domain
 - (C) $y = f(x)$ has exactly one point of inflexion
 - (D) $y = f(x)$ is differentiable everywhere in its domain.
- 6 The values of the real numbers p and q that makes $1-i$ a root of the equation $z^3 + pz + q = 0$ are:
- (A) $p = -2$ and $q = -4$
 - (B) $p = -2$ and $q = 4$
 - (C) $p = 2$ and $q = -4$
 - (D) $p = 2$ and $q = 4$

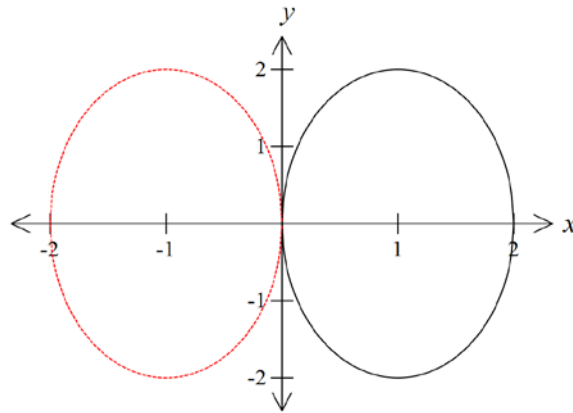
- 7 Let $P(x)$ be a polynomial of degree $n > 0$ such that $P(x) = (x - \alpha)^r Q(x)$, where $r \geq 2$ and α is a real number. $Q(x)$ is a polynomial with real coefficients of degree $q > 0$. Which of the following is the incorrect statement?
- (A) $n \leq r + q$
 - (B) $P(x)$ changes sign around the root $x = \alpha$
 - (C) Let N_r be the number of real roots of $P(x)$ and N_c the number of complex roots of $P(x)$. Then $r \leq N_r \leq n$ and $0 \leq N_c \leq q$.
 - (D) Roots of $P(x)$ are conjugate one another.

- 8 The eccentricity of the ellipse with the equation $\frac{x^2}{4} + \frac{y^2}{3} = 1$ is:
- (A) $\frac{1}{4}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{3}{4}$
 - (D) $\frac{9}{16}$

- 9 A particle of mass m falls from rest under gravity and the resistance to its motion is mkv , where v is its speed and k is a positive constant. Which of the following is the correct expression for the velocity?

- (A) $v = \frac{g}{k}(1 - e^{-kt})$
- (B) $v = \frac{g}{k}(1 + e^{-kt})$
- (C) $v = \frac{g}{k}(1 - e^{kt})$
- (D) $v = \frac{g}{k}(1 + e^{kt})$

- 10 The region enclosed by the ellipse $(x-1)^2 + \frac{y^2}{4} = 1$ is rotated about the y axis to form a solid.



What is the correct expression for volume of this solid using the method of slicing?

- (A) $V = \int_{-2}^2 \pi \sqrt{1-y^2} dy$
- (B) $V = \int_{-2}^2 2\pi \sqrt{1-y^2} dy$
- (C) $V = \int_{-2}^2 \pi \sqrt{4-y^2} dy$
- (D) $V = \int_{-2}^2 2\pi \sqrt{4-y^2} dy$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) The complex number z is given by $z = -1 + \sqrt{3}i$
- (i) Show that $z^2 = 2\bar{z}$ 2
- (ii) Evaluate $|z|$ and $\text{Arg}(z)$ 2
- (b) Calculate the product of the roots of the following equation in the form $a + ib$ 2
- $$(3 + 2i)z^2 - (1 - 2i)z + (6 - i) = 0$$
- (c) Find the complex square roots of $7 - 6\sqrt{2}i$ giving your answers in the form $x + iy$, 3
- where x and y are real.
- (d) (i) Express $z = \sqrt{3} + i$ in modulus/argument form. 3
- (ii) Show that $z^7 + 64z = 0$
- (e) The points A, B, C, D on the Argand diagram represent the complex numbers a, b, c, d 3
- respectively. If $a + c = b + d$ and $a - c = i(b - d)$ find what type of quadrilateral $ABCD$ is.

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int_0^3 \frac{\sqrt{x}}{1+x} dx$ (Hint: let $u^2 = x$) 3

(b) By using a suitable trigonometric substitution show that $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c$ 2

where c is some constant

(c) (i) Find real numbers A, B and C such that $\int \frac{x+6}{(x+1)(x^2+9)} dx \equiv \int \frac{A}{x+1} + \int \frac{Bx+C}{x^2+9}$ 2

(ii) Hence, find $\int \frac{x+6}{(x+1)(x^2+9)} dx$ 2

(d) Using the substitution $t = \tan \frac{x}{2}$, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{5+3\cos x}$ 3

(e) For $n \geq 0$ let, 3

$$I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$$

Show that for $n \geq 2$, $I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1)I_{n-2}$

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The area bounded by the curve $y = 12x - x^2$, the x axis, $x = 2$ and $x = 10$ is rotated about the y axis to form a solid. By using the method of cylindrical shells calculate the volume of the solid. **4**
- (b) The base of a solid is the segment of the parabola $x^2 = 4y$ cut off by the line $y = 2$. Each cross section perpendicular to the y axis is a right angled isosceles triangle with the hypotenuse in the base of the solid. Find the volume of the solid. **3**
- (c) Consider the function $f(x) = \frac{e^x - 1}{e^x + 1}$
- (i) Show that $f(x)$ is an odd function **1**
 - (ii) Show that the function is always increasing **2**
 - (iii) Find $f'(0)$ **1**
 - (iv) Discuss the behaviour of $f(x)$ as $x \rightarrow \pm \infty$ **2**
 - (v) Sketch the graph of $y = f(x)$ **2**

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) For the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$

- (i) Find the eccentricity 1
- (ii) Find the coordinates of the foci S and S' 1
- (iii) Find the coordinates of the directrices. 1
- (iv) Sketch the curve showing foci and directrices. 1

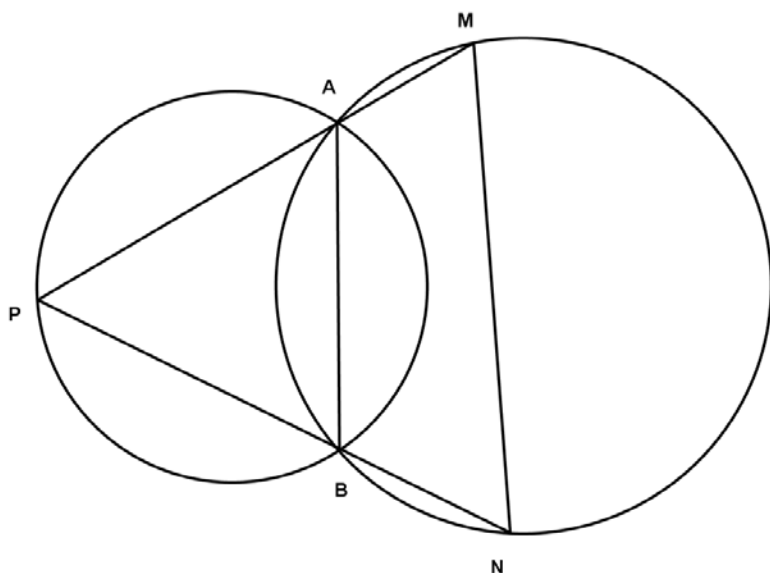
P is an arbitrary point on this ellipse

- (v) Prove that the sum of the distances SP and $S'P$ is independent of P 2
- (b) (i) Find the slope of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point 2

$P(a \cos \theta, b \sin \theta)$

- (ii) Hence show that the equation of this tangent is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. 2
- (iii) If the point $P(a \cos \theta, b \sin \theta)$ is on the ellipse in quadrant one, find the minimum area of the triangle made by this tangent and the coordinate axes. 2

- (c) Two fixed circles intersect at AB . P is a variable point on one circle.



Copy or trace the diagram into your writing booklet.

- (i) Let $\angle APB = \theta$, explain why θ is a constant. 1
- (ii) Prove that MN is of a constant length. 2

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) A body of unit mass falls under gravity through a resisted medium. The body falls from rest. The resistance to its motion is $\frac{1}{100}v^2$ Newtons where v metres per second is the speed of the body when it has fallen a distance x metres.

(i) Show that the equation of motion of the body is $\ddot{x} = g - \frac{1}{100}v^2$, where g is the magnitude of the acceleration due to gravity. (Note: Draw a diagram!) 1

(ii) Show that the terminal speed, V_T , is given by $V_T = 10\sqrt{g}$ 1

(iii) Show that $V^2 = V_T^2 \left(1 - e^{-\frac{x}{50}}\right)$ 3

- (b) $x^3 - 6x^2 + 9x + k = 0$ has two equal roots.

(i) Show that $k = -4$ is a possible value for k 2

(ii) Solve $x^3 - 6x^2 + 9x - 4 = 0$ 2

- (c) α, β , and γ are the roots of $x^3 + px^2 + qx + r = 0$

Find

(i) $\alpha^2 + \beta^2 + \gamma^2$ 1

(ii) $\alpha^3 + \beta^3 + \gamma^3$ 2

- (d)

(i) Prove that $x^2 + x + 1 \geq 0$ for all real x 1

(ii) Hence or otherwise, prove that $a^2 + ab + b^2 \geq 0$ 2

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) Let $w = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$
- (i) Show that w^k is a solution of $z^7 - 1 = 0$, where k is an integer. 2
- (ii) Prove that $w + w^2 + w^3 + w^4 + w^5 + w^6 = -1$ 2
- (iii) Hence show that $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$ 2

- (b) The Bernoulli polynomials $B_n(x)$ are defined by $B_0(x) = 1$ and for $n = 1, 2, 3, \dots$,

$$\frac{dB_n(x)}{dx} = nB_{n-1}(x), \text{ and}$$

$$\int_0^1 B_n(x) dx = 0$$

Thus

$$B_1(x) = x - \frac{1}{2}$$

$$B_2(x) = x^2 - x + \frac{1}{6}$$

$$B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$$

- (i) Show that $B_4(x) = x^2(x-1)^2 - \frac{1}{30}$ 2
- (ii) Show that, for $n \geq 2$, $B_n(1) - B_n(0) = 0$ 1
- (iii) Show by mathematical induction, that for $n \geq 1$ 3
- $$B_n(x+1) - B_n(x) = n^{-1}x^{n-1},$$
- (iv) Hence show that for $n \geq 1$ and any positive integer k 2
- $$n \sum_{m=0}^k m^{n-1} = B_n(k+1) - B_n(0)$$
- (v) Hence deduce that $\sum_{m=0}^{135} m^4 = 9134962308$ 1

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note: $\ln x = \log_e x$, $x > 0$

SECTION I

MULTIPLE CHOICE

(1) B

$$\begin{aligned} (2) \quad \frac{5}{i(-2+i)} &= \frac{5}{-2i+i^2} \\ &= \frac{5}{-1-2i} \times \frac{-1+2i}{-1+2i} \\ &= \frac{-5+10i}{5} \\ &= -1+2i \end{aligned}$$

B

(3) C

(4) D

(5) C

(6) Roots $= (1-i), (1+i), \gamma$

$$\sum \alpha = -\frac{b}{a} = 0$$

$$\sum \alpha\beta = \frac{c}{a} = p$$

$$\sum \alpha\beta\gamma = -\frac{d}{a} = -q$$

$$\sum \alpha = (1-i) + (1+i) + \gamma = 0$$

$$\Rightarrow 2 + \gamma = 0$$

$$\Rightarrow \gamma = -2$$

$$\sum \alpha\beta\gamma = (1-i)(1+i)(-2)$$

$$2(-2) = -q$$

$$\therefore q = 4$$

$$\sum \alpha\beta = (1-i)(1+i) + -2(1-i) + -2(1+i)$$

$$2 - 2 + 2i - 2 - 2i$$

$$-2 = p$$

$$\therefore p = -2 \quad q = 4$$

B

(7) D

$$(8) \quad b^2 = a^2(1-e^2)$$

$$3 = 4(1-e^2)$$

$$\frac{3}{4} = 1-e^2$$

$$e^2 = 1 - \frac{3}{4}$$

$$e = \frac{1}{2}$$

B

(9) $\downarrow mg \quad \uparrow mkv$

$$\ddot{x} = g - kv$$

$$\frac{dv}{dt} = g - kv$$

$$\frac{dt}{dv} = -\frac{1}{k} \cdot \frac{-k}{g-kv}$$

$$\textcircled{1} - t = -\frac{1}{k} \ln(g-kv) + C$$

$$\textcircled{2} - 0 = -\frac{1}{k} \ln(g) + C \quad t=0 \quad v=0$$

$$\textcircled{1} - \textcircled{2} \quad t = -\frac{1}{k} \ln\left(\frac{g-kv}{g}\right)$$

$$-kt = \ln\left(\frac{g-kv}{g}\right)$$

$$e^{-kt} = \frac{g-kv}{g}$$

$$ge^{-kt} = g - kv$$

$$kv = g - ge^{-kt}$$

$$v = \frac{g}{k}(1 - e^{-kt})$$

(A)

(10)

$$\pi \int x_2^2 - x_1^2 dy$$

$$V = \pi \int_{-2}^2 (x_2^2 - x_1^2) dy$$

$$(x-1)^2 + \frac{y^2}{4} = 1$$

$$x^2 - 2x + 1 + \frac{y^2}{4} = 1$$

$$x^2 - 2x + \frac{y^2}{4} = 0$$

$$\Sigma \alpha = -2 \quad x_1 + x_2 = -2$$

$$\Sigma \alpha \beta = \frac{y^2}{4} \quad x_1 x_2 = \frac{y^2}{4}$$

$$x_2^2 - x_1^2 = (x_2 - x_1)(x_2 + x_1)$$

$$\begin{aligned} x_2 - x_1 &= \sqrt{(x_2 + x_1)^2} \\ &= \sqrt{(x_2 + x_1)^2 - 4x_1 x_2} \\ &= \sqrt{(-2)^2 - 4\left(\frac{y^2}{4}\right)} \end{aligned}$$

$$x_2 - x_1 = \sqrt{4 - y^2}$$

$$\begin{aligned} \therefore x_2^2 - x_1^2 &= \sqrt{4 - y^2} \times 2 \\ &= 2\sqrt{4 - y^2} \end{aligned}$$

$$\int_{-2}^2 2\pi \sqrt{4 - y^2} dy$$

(D)

SECTION II

(11)

$$a) i) \quad z = -1 + \sqrt{3}i$$

$$z^2 = (-1 + \sqrt{3}i)^2$$

$$= 1 - 2\sqrt{3}i - 3$$

$$= -2 - 2\sqrt{3}i$$

$$\bar{z} = -1 - \sqrt{3}i$$

$$2\bar{z} = -2 - 2\sqrt{3}i$$

$$= z^2$$

$$\begin{aligned} (ii) \quad |z| &= \sqrt{(-1)^2 + (\sqrt{3})^2} \\ &= \sqrt{1+3} \\ &= 2 \end{aligned}$$



$$\text{Arg } z = 120 \text{ or } \frac{2\pi}{3}$$

$$b) \quad \alpha\beta = \frac{c}{a}$$

$$= \frac{6-i}{3+2i} \times \frac{3+2i}{3-2i}$$

$$= \frac{18 - 12i - 3i - 2}{9 + 4}$$

$$= \frac{16 - 15i}{13}$$

$$= \frac{16}{13} - \frac{15}{13}i$$

$$c) \cdot z^2 = 7 - 6\sqrt{2}i$$

$$x^2 - y^2 + 2ixy = 7 - 6\sqrt{2}i$$

$$x^2 - y^2 = 7$$

$$2xy = -6\sqrt{2} \quad *$$

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2$$

$$= 49 + 72$$

$$(x^2 + y^2)^2 = 121$$

$$x^2 + y^2 = 11 \quad - (1)$$

$$x^2 - y^2 = 7 \quad - (2)$$

$$(1) + (2) \quad 2x^2 = 18$$

$$(1) - (2) \quad 2y^2 = 4$$

$$x^2 = 9 \quad \therefore x = \pm 3$$

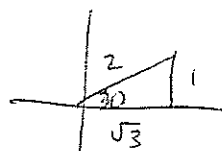
$$y^2 = 2 \quad \therefore y = \pm \sqrt{2}$$

check in *

$$2xy = -6\sqrt{2}$$

$$\therefore \pm (3 - 2i)$$

$$d) i) \quad z = \sqrt{3} + i$$



$$z = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$(ii) \quad z^7 = 2^7 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$$

$$= 128 \left(\cos -\frac{5\pi}{6} + i \sin -\frac{5\pi}{6} \right)$$

$$= 128 \left(-\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$$

$$= -128 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$= -64 \cdot 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$= -64z$$

$$(e) \quad a+c = b+d$$

$$\frac{a+c}{2} = \frac{b+d}{2}$$

$\therefore$ midpoint AC = midpoint BD

$\therefore$ diagonals AC, BD bisect each other

$$a-c = i(b-d)$$

$\therefore$ diagonal AC and BD

are equal and perpendicular

$\therefore$ ABCD is a square

$$(12) a) \int_0^3 \frac{\sqrt{x}}{1+x} dx$$

$$\text{Let } u^2 = x$$

$$2u \frac{du}{dx} = 1$$

$$\frac{du}{dx} = \frac{1}{2u} \quad 2u du = dx$$

$$dx =$$

$$= \int \frac{u}{1+u^2} \cdot 2u du$$

$$= \int \frac{2u^2}{1+u^2} du$$

$$= 2 \int 1 - \frac{1}{1+u^2}$$

$$= 2 \left[u - \tan^{-1}(u) \right]_0^{\sqrt{3}}$$

$$= 2 \left[\sqrt{3} - \frac{\pi}{3} \right]$$

$$= \frac{6\sqrt{3} - 2\pi}{3}$$

$$b) \int \frac{dx}{\sqrt{1-x^2}}$$

$$\text{Let } x = \sin \theta$$

$$\frac{dx}{d\theta} = \cos \theta$$

$$dx = \cos \theta d\theta$$

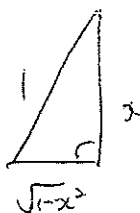
$$= \int \frac{1}{\sqrt{1-\sin^2 \theta}} \cos \theta d\theta$$

$$= \int \frac{\cos \theta}{\cos \theta} d\theta$$

$$= \int 1 d\theta$$

$$= \theta + C$$

$$= \sin^{-1} x + C$$



$$c(i) \int \frac{x+6}{(x+1)(x^2+9)} dx$$

$$(x^2+9)A + (x+1)(Bx+C) = x+6$$

$$\text{Let } x = -1$$

$$10A = 5$$

$$A = \frac{1}{2}$$

$$\text{Let } x = 0$$

$$9\left(\frac{1}{2}\right) + 1(C) = 6$$

$$9 + 2C = 12$$

$$2C = 3$$

$$C = \frac{3}{2}$$

$$\text{Let } x = 1$$

$$\frac{10}{2} + 2\left(B + \frac{3}{2}\right) = 7$$

$$2\left(B + \frac{3}{2}\right) = 2$$

$$B + \frac{3}{2} = 1$$

$$B = -\frac{1}{2}$$

$$A = \frac{1}{2} \quad B = -\frac{1}{2} \quad C = \frac{3}{2}$$

$$(ii) \int \frac{x+6}{(x+1)(x^2+9)} dx$$

$$\frac{1}{2} \int \frac{1}{x+1} + \int \frac{-\frac{1}{2}x + \frac{3}{2}}{x^2+9}$$

$$= \frac{1}{2} \ln(x+1) + -\frac{1}{4} \int \frac{2x-6}{x^2+9}$$

$$= \frac{1}{2} \ln(x+1) + -\frac{1}{4} \left[\frac{2x}{x^2+9} - 6 \int \frac{1}{x^2+9} \right]$$

$$= \frac{1}{2} \ln(x+1) - \frac{1}{4} \ln(x^2+9) + \frac{3}{2} \times \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C$$

$$= \frac{1}{2} \ln(x+1) - \frac{1}{4} \ln(x^2+9) + \frac{1}{2} \tan^{-1}\left(\frac{x}{3}\right) + C$$

(d) Let $t = \tan \frac{x}{2}$

$$\therefore dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$= \frac{1+t^2}{2} dx \quad \begin{array}{l} x=0 \quad t=0 \\ x=\frac{\pi}{2} \quad t=1 \end{array}$$

$$dx = \frac{2 dt}{1+t^2}$$

$$\int \frac{dx}{5+3\cos x} = \int_0^1 \frac{2 dt}{5+3\left(\frac{1-t^2}{1+t^2}\right)}$$

$$= \int_0^1 \frac{2 dt}{5+5t^2+3-3t^2}$$

$$= \int_0^1 \frac{2 dt}{8+2t^2}$$

$$= \int_0^1 \frac{1}{4+t^2}$$

$$= \frac{1}{2} \left[\tan^{-1} \frac{t}{2} \right]_0^1$$

$$= \frac{1}{2} \left[\tan^{-1} \frac{1}{2} \right]$$

(e) $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$

$$u = x^n \quad u' = nx^{n-1}$$

$$v' = \sin x \quad v = -\cos x$$

$$= \left[-x^n \cos x \right]_0^{\frac{\pi}{2}} + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x dx$$

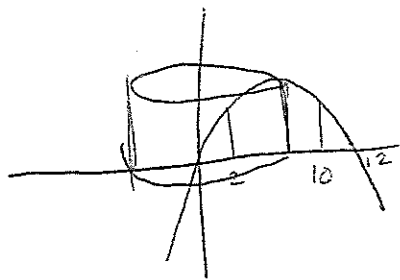
$$= n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x dx$$

$$\int_0^{\frac{\pi}{2}} x^{n-1} \cos x dx = \left[x^{n-1} \sin x \right]_0^{\frac{\pi}{2}} - (n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x dx$$

$$= \left(\frac{\pi}{2} \right)^{n-1} - (n-1) I_{n-2}$$

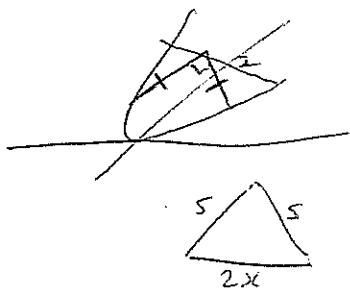
$$\therefore I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

(13)
a)



$$\begin{aligned}
 V &= \int 2\pi xy \, dx \\
 &= 2\pi \int_2^{10} x(12x - x^2) \, dx \\
 &= 2\pi \int_2^{10} 12x^2 - x^3 \, dx \\
 &= 2\pi \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_2^{10} \\
 &= 2\pi \left[\left(\frac{4000}{3} - \frac{10000}{4} \right) - \left(\frac{32}{3} - 4 \right) \right] \\
 &= 2\pi [1500 - 28] \\
 &= 2944\pi \text{ units}^2
 \end{aligned}$$

b)



$$2s^2 = 4x^2$$

$$s^2 = 2x^2$$

$$s = x\sqrt{2}$$

area of triangle is $\frac{1}{2}b \times h$

$$\frac{1}{2}s^2$$

$$= x^2$$

$$= 4y$$

$$\begin{aligned}
 V &= \int_0^2 4y \, dy \\
 &= \left[2y^2 \right]_0^2 = 8 \text{ cubic units}
 \end{aligned}$$

$$(c)(i) \quad f(x) = \frac{e^x - 1}{e^x + 1}$$

$$f(-x) = \frac{e^{-x} - 1}{e^{-x} + 1} \times \frac{e^x}{e^x}$$

$$= \frac{1 - e^x}{1 + e^x}$$

$$= -\frac{e^x - 1}{e^x + 1}$$

$$= -f(x)$$

$$(ii) \quad f'(x) = \frac{e^x(e^x + 1) - (e^x - 1)e^x}{(e^x + 1)^2}$$

$$= \frac{2e^x}{(e^x + 1)^2}$$

but $e^x > 0$

$$(e^x + 1)^2 > 1$$

$$\therefore f'(x) > 0$$

$\therefore f(x)$ is always increasing

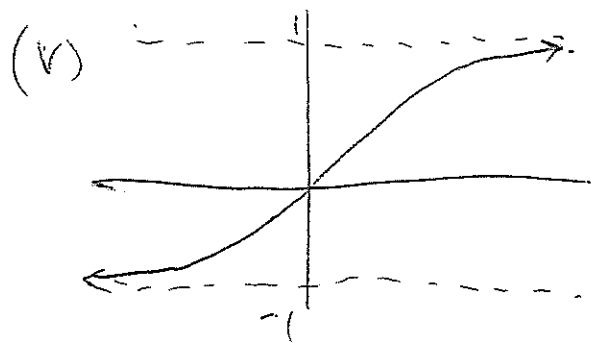
$$(iii) \quad f'(0) = \frac{2e^0}{(e^0 + 1)^2}$$

$$= \frac{2}{2^2} = \frac{1}{2}$$

$$(iv) \quad f(x) = \frac{e^x - 1}{e^x + 1}$$

$$= \frac{1 - e^{-x}}{1 + e^{-x}} \rightarrow 1 \text{ as } x \rightarrow \infty$$

$$f(x) = \frac{e^x - 1}{e^x + 1} \rightarrow -1 \text{ as } x \rightarrow -\infty$$



(14) $\frac{x^2}{4} + \frac{y^2}{3} = 1$ $a=2$ $b=\sqrt{3}$

a)(i) $b^2 = a^2(1-e^2)$

$$3 = 4(1-e^2)$$

$$\frac{3}{4} = 1-e^2$$

$$e^2 = \frac{1}{4}$$

$$e = \frac{1}{2}$$

(ii) $S = (ae, 0)$

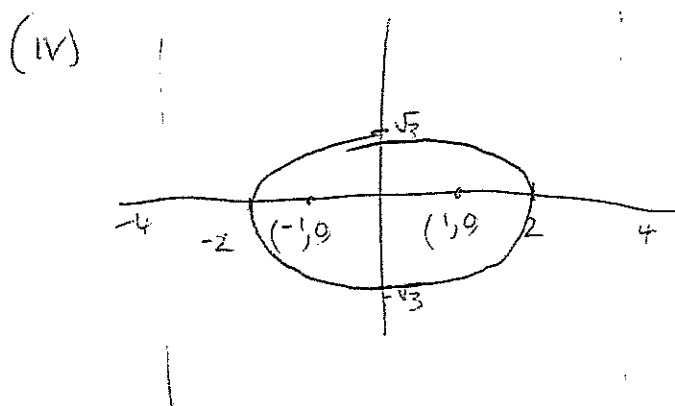
$$= (1, 0)$$

$$S' = (-ae, 0)$$

$$= (-1, 0)$$

(iii) $x = \pm \frac{a}{e}$

$$\Rightarrow x = \pm 4$$



(v) $PS + PS' = e PD + e PD'$

Since $e = \frac{PS}{PD} = \frac{PS'}{PD'}$

$$= e(PD + PD')$$

$$= 8e$$

$\therefore$ independent of P

(b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$b^2 x^2 + a^2 y^2 = a^2 b^2$$

$$2b^2 x + 2a^2 y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

at $P(a \cos \theta, b \sin \theta)$

$$\frac{dy}{dx} = \frac{-ab^2 \cos \theta}{a^2 b \sin \theta}$$

$$= \frac{-b \cos \theta}{a \sin \theta}$$

(i) $y - b \sin \theta = -\frac{b}{a} \frac{\cos \theta}{\sin \theta} (x - a \cos \theta)$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

(ii) when $x=0$ $y = \frac{b}{\sin \theta}$

when $y=0$ $x = \frac{a}{\cos \theta}$

so $A(0, \frac{b}{\sin \theta})$

$B(\frac{a}{\cos \theta}, 0)$

Area of $\triangle AOB$

$= \frac{1}{2} (OB \cdot OA)$

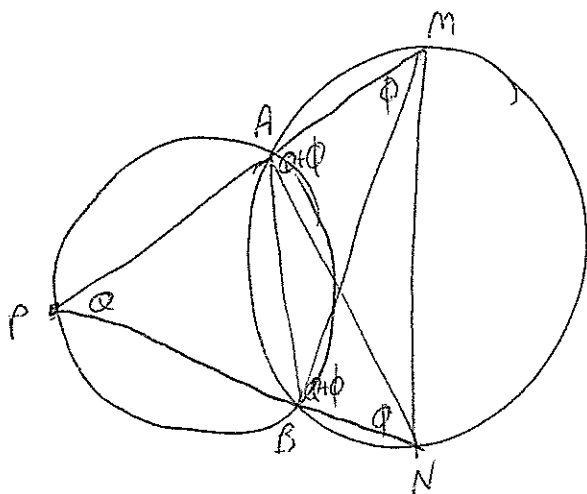
$= \frac{ab}{2 \sin \theta \cos \theta}$

$= \frac{ab}{\sin 2\theta}$

min when $\sin 2\theta$ max

$\sin 2\theta = 1$ ie when $\theta = \frac{\pi}{4}$

(c)



(i) since AB is a fixed chord

$\therefore \theta$ is a constant since angle subtended from same chord AB are equal.

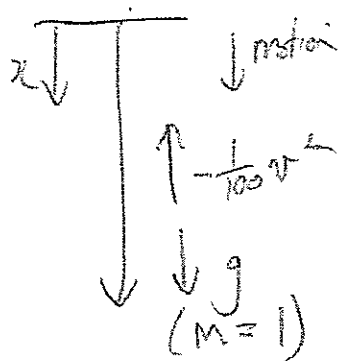
(ii) $\angle \phi$ is fixed (angle subtended by same chord AB,

$\angle MAN = \angle MBN = \theta + \phi$ (both are exterior angles to $\triangle PAN$ and $\triangle PBM$ respectively
= constant.

$\therefore MN$ is a constant
since fixed circumference angles subtends fixed chords in the same circle.

15

(a)



$$F = mg - \frac{1}{100} v^2$$

$$\ddot{x} = g - \frac{1}{100m} v^2$$

$m=1$

$$\ddot{x} = g - \frac{1}{100} v^2$$

(i) when terminal

$$\sum F = 0$$

$$\therefore g - \frac{1}{100} V_T^2 = 0$$

$$\begin{aligned} V_T^2 &= 100g \\ &= \sqrt{100g} \\ &= 10\sqrt{g} \end{aligned}$$

$$(ii) \quad v \frac{dv}{dx} = g - \frac{1}{100} v^2$$

$$\int \frac{v dv}{g - \frac{1}{100} v^2} = \int dx$$

$$\begin{aligned} \text{L.H.S.} &= \int \frac{100v}{100g - v^2} \\ &= \int \frac{100v}{V_T^2 - v^2} \end{aligned}$$

$$= -\frac{100}{2} \ln(V_T^2 - v^2)$$

$$-50 \ln(V_T^2 - v^2) = x + C$$

$$V_T^2 - v^2 = A e^{-\frac{x}{50}}$$

$$v^2 = V_T^2 - A e^{-\frac{x}{50}}$$

when $x=0$ $v=0$

$$0 = V_T^2 - A$$

$$A = V_T^2$$

$$v^2 = V_T^2 (1 - e^{-\frac{x}{50}})$$

b) $P(x) = 3x^2 - 12x + 9$

$$3(x^2 - 4x + 3) = 0$$

$$3(x-3)(x-1) = 0$$

$$x=1 \text{ or } x=3$$

P(1) $P(1) = 0$

~~$$3x^2$$~~

$$1 - 6 + 9 + k = 0$$

$$k = -4$$

(ii) $x^3 - 6x^2 + 9x - 4 = 0$

$$(x-1)^2 = x^2 - 2x + 1$$

$$\begin{array}{r} x^3 - 6x^2 + 9x - 4 \\ (x^2 - 2x + 1) \overline{) x^3 - 6x^2 + 9x - 4} \\ \underline{x^3 - 2x^2 - 4} \\ -4x^2 + 9x - 4 \\ \underline{-4x^2 + 9x - 4} \\ 0 \end{array}$$

$$7x = 4$$

$$\therefore x = 1, 1, 4$$

$$\begin{aligned}
 \text{c(i)} \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\
 &= -p^2 - 2q \\
 &= p^2 - 2q.
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \alpha^3 &= -p\alpha^2 - q\alpha - r \\
 \beta^3 &= -p\beta^2 - q\beta - r \\
 \gamma^3 &= -p\gamma^2 - q\gamma - r \\
 \alpha^3 + \beta^3 + \gamma^3 &= -p(\alpha^2 + \beta^2 + \gamma^2) - q(\alpha + \beta + \gamma) - 3r \\
 &= -p(p^2 - 2q) - q(-p) - 3r \\
 &= -p^3 + 3pq - 3r.
 \end{aligned}$$

$$\text{d)} \quad x^2 + x + 1 \geq 0.$$

$$\Delta = b^2 - 4ac$$

$$1 - 4 = -3 < 0. \quad \therefore \text{always true.}$$

$$\begin{aligned}
 \text{(ii)} \quad x^2 + ax + b^2 &= b^2 \left(\frac{x^2}{b^2} + \frac{ax}{b^2} + \frac{b^2}{b^2} \right) \\
 &= b^2 \left(\left(\frac{x}{b} \right)^2 + \left(\frac{a}{b} \right) \left(\frac{x}{b} \right) + 1 \right) \\
 &= b^2 (x^2 + x + 1) \quad \text{if } x = \frac{a}{b} \\
 &\geq 0.
 \end{aligned}$$

(16) a(i)

$$w = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$$

$$w^k = \cos \frac{2k\pi}{7} + i \sin \frac{2k\pi}{7}$$

$$\begin{aligned} \therefore z^7 - 1 &= (w^k)^7 - 1 \\ &= (\cos 2k\pi + i \sin 2k\pi) - 1 \\ &= (1 + 0) - 1 = 0 \end{aligned}$$

(ii) $w + w^2 + w^3 + \dots + w^6$

$$\begin{aligned} \text{G.P} &= \frac{a(1-r^n)}{1-r} \\ &= \frac{w(1-w^6)}{1-w} \end{aligned}$$

$$= \frac{w - w^7}{1-w}$$

$$= \frac{w-1}{1-w}$$

$$= -1$$

(iii) $w + w^6 = 2 \cos \frac{2\pi}{7}$
 $w^2 + w^5 = 2 \cos \frac{4\pi}{7}$
 $w^3 + w^4 = 2 \cos \frac{6\pi}{7}$

$$2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{6\pi}{7} = -1$$

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$$

(b)

(i) $B_4(x) = 4 \int B_3(x) dx$
 $= \int 4x^3 - 6x^2 + 2x dx$
 $= x^4 - 2x^3 + x^2 + C$

Now $\int_0^1 x^4 - 2x^3 + x^2 + C dx = 0$
 $\therefore \left[\frac{1}{5}x^5 - \frac{1}{2}x^4 + \frac{1}{3}x^3 + Cx \right]_0^1 = 0$

$$\begin{aligned} \therefore C &= \frac{1}{5} - \frac{1}{2} + \frac{1}{3} \\ &= -\frac{1}{30} \end{aligned}$$

$$\begin{aligned} \therefore B_4(x) &= x^4 - 2x^3 + x^2 - \frac{1}{30} \\ &= x^2(x-1)^2 - \frac{1}{30} \end{aligned}$$

(ii) $B_n(1) - B_n(0)$

$$= \int_0^1 B_{n-1}(x) dx$$

$$= 0 \text{ by definition}$$

ie $B_n(1) - B_n(0) = 0$

If $n=1$: $\int_0^1 B_0(x) dx = \int_0^1 dx = 1$
 $= 0$

(iii) Let $\{B_n\}$ by the statement

that $B_n(x+1) - B_n(x) = nx^{n-1}$
 for some positive integer n .

Now $B_1(x+1) - B_1(x)$
 $= x+1 - \frac{1}{2} - (x - \frac{1}{2})$
 $= 1$
 $= 1 \cdot x^{1-1}$

hence $S(1)$ true

Let k be some positive integer for which $S(k)$ true

ie $B_k(x+1) - B_k(x) = kx^{k-1}$

Consider

$$\frac{d}{dx} [B_{k+1}(x+1) - B_{k+1}(x)]$$

$$= (k+1) B_k(x+1) - (k+1) B_k(x)$$

$$= (k+1) [B_k(x+1) - B_k(x)]$$

$$= (k+1) k x^{k-1} \text{ by assumption}$$

$$B_{k+1}(x) - B_k(x) = \frac{1}{k+1} x^k$$

$$(iv) B_n(1) - B_n(0) = n \cdot 0^{n-1}$$

$$B_n(2) - B_n(1) = n \cdot 1^{n-1}$$

$$B_n(3) - B_n(2) = n \cdot 2^{n-1}$$

$$\vdots$$

$$B_n(k) - B_n(k-1) = n(k-1)^{n-1}$$

$$B_n(k+1) - B_n(k) = n k^{n-1}$$

Sum of L.H.S -

$$B_n(k+1) - B_n(0)$$

Sum of R.H.S

$$n(0^{n-1} + 1^{n-1} + \dots + k^{n-1})$$

$$= n \sum_{m=0}^k m^{n-1}$$

$$(v) \sum_{m=0}^{135} m^4 = \frac{1}{5} (B_5(136) - B_5(0))$$

$$B_5(x) = 5 \int B_0(x) dx$$

$$B_5(x) = x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 - \frac{x}{6} + C$$

$$= \frac{1}{5} (136^5 - \frac{5}{2} \times 136^4 + \frac{5}{3} \times 136^3 - \frac{1}{6} \times 136 + C - C)$$

$$= 9134962308$$